

Econ 30065: Microeconomics

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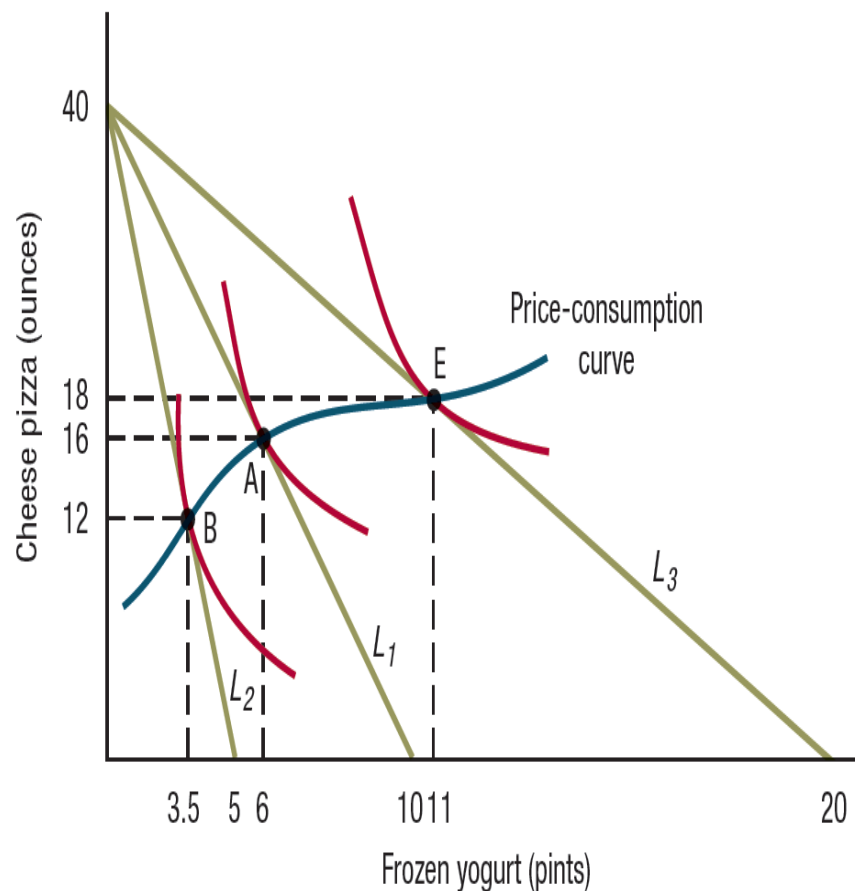
L8

Price-Consumption Curve

- Past class: best choice **given** prices and income.
- If P_x or P_y changes: change in affordable bundles set
- The **price-consumption curve** shows how the optimal consumption bundle changes as the **price of one good changes**, holding everything else fixed

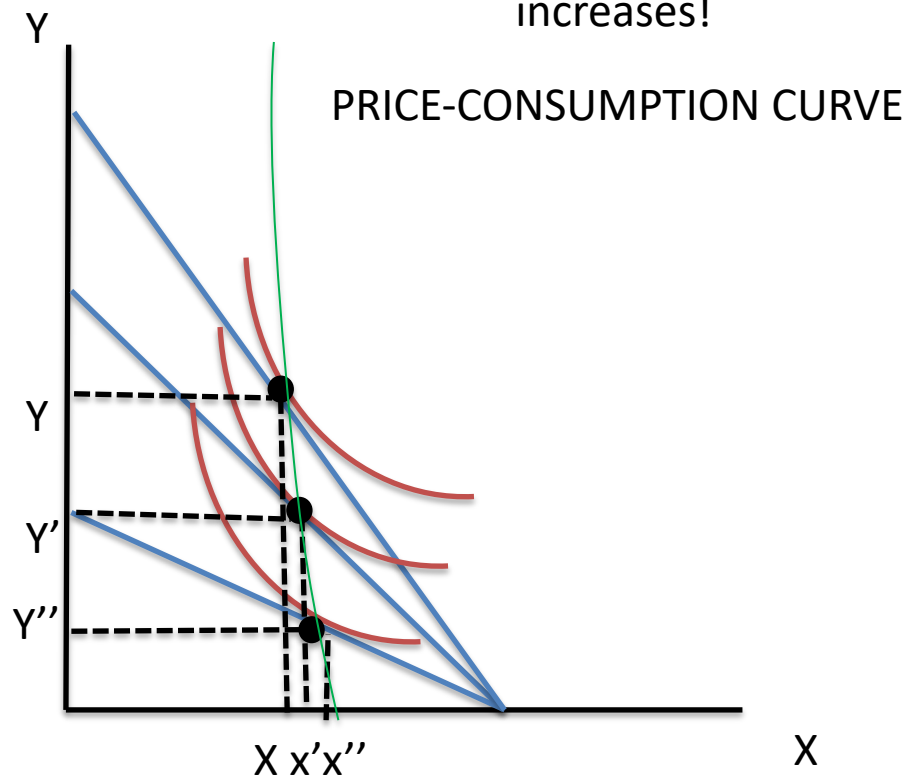
Price-consumption curve when P_x changes

- It shows the chosen bundle for each possible price of frozen yoghurt (1,2,0.50\$), given the price of pizza (at 0.25\$) and income (at 10\$)
- Quantity chosen of Frozen yogurt decreases as its price increases!



Price-consumption curve when P_Y changes

- It shows the chosen bundle for each possible price of Y (given the price of X and income)
- Quantity chosen of Y decreases as its price increases!



Individual demand curves

➤ Price-consumption curve includes all the information needed to plot an individual's demand curve

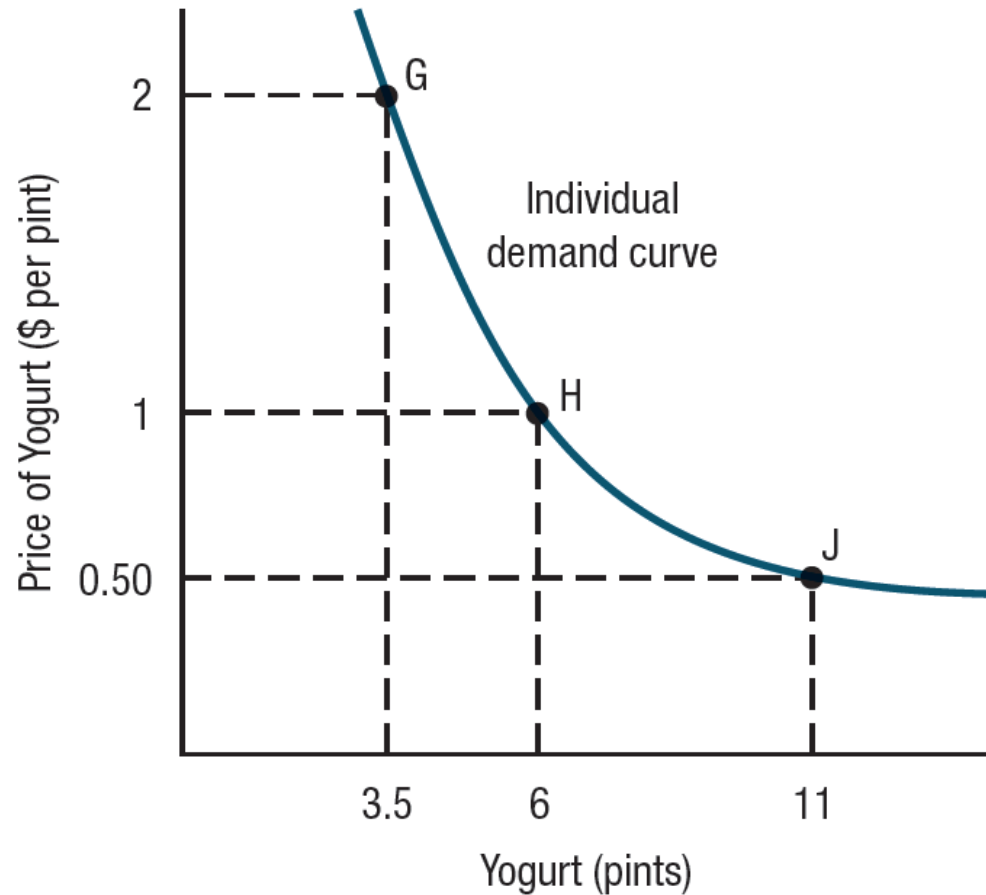
➤ An **individual demand curve**:

$$\textit{Quantity Demanded} = D(\text{Price, Other factors})$$

➤ Describes the relationship between the prices of a good and the amount a consumer purchases of that good

➤ Holds everything else fixed

Individual demand curve for frozen yoghurt



Individual demand curve for goods

- From price (of x)-consumption curve find demand curve for good x
- From price (of y)-consumption curve find demand curve for good y
- *Law of the demand: if the price of a good increases, quantity demanded of that good decreases.*

Back to our example:

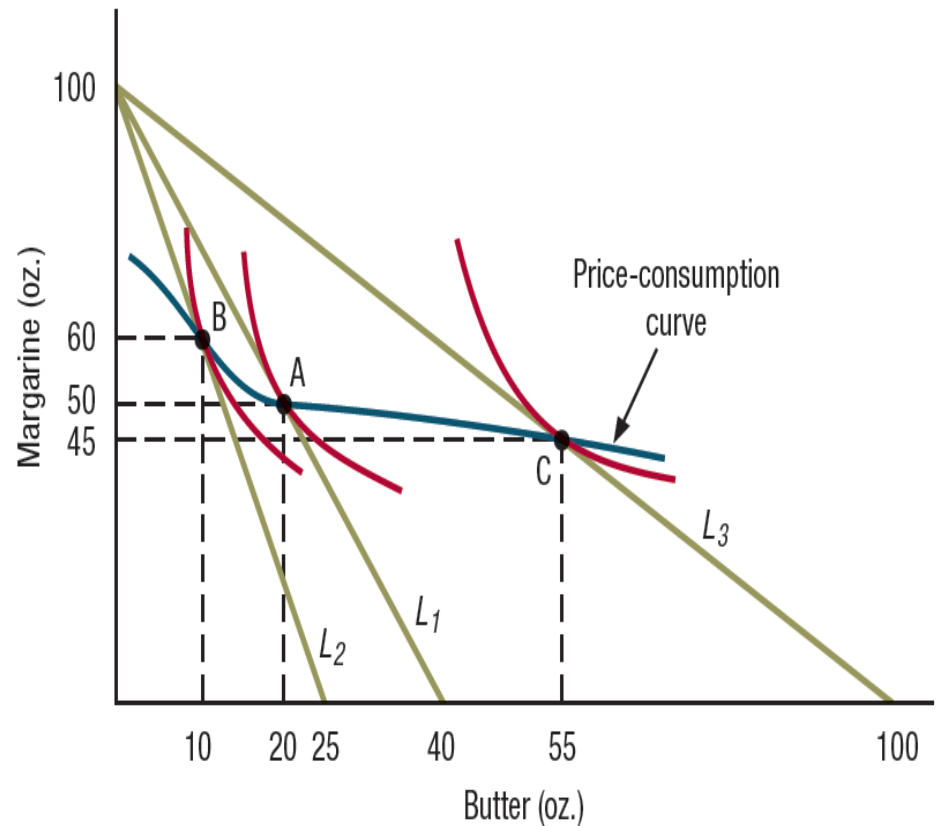
$$U(X, Y) = XY^4$$

- Remember: we had $MRS_{xy} = (1/4)(y/x)$.
- Derive analytically the demand function for x and y as a function of p_x (still assuming that $p_y = 1$ € and $I = 300$ €).

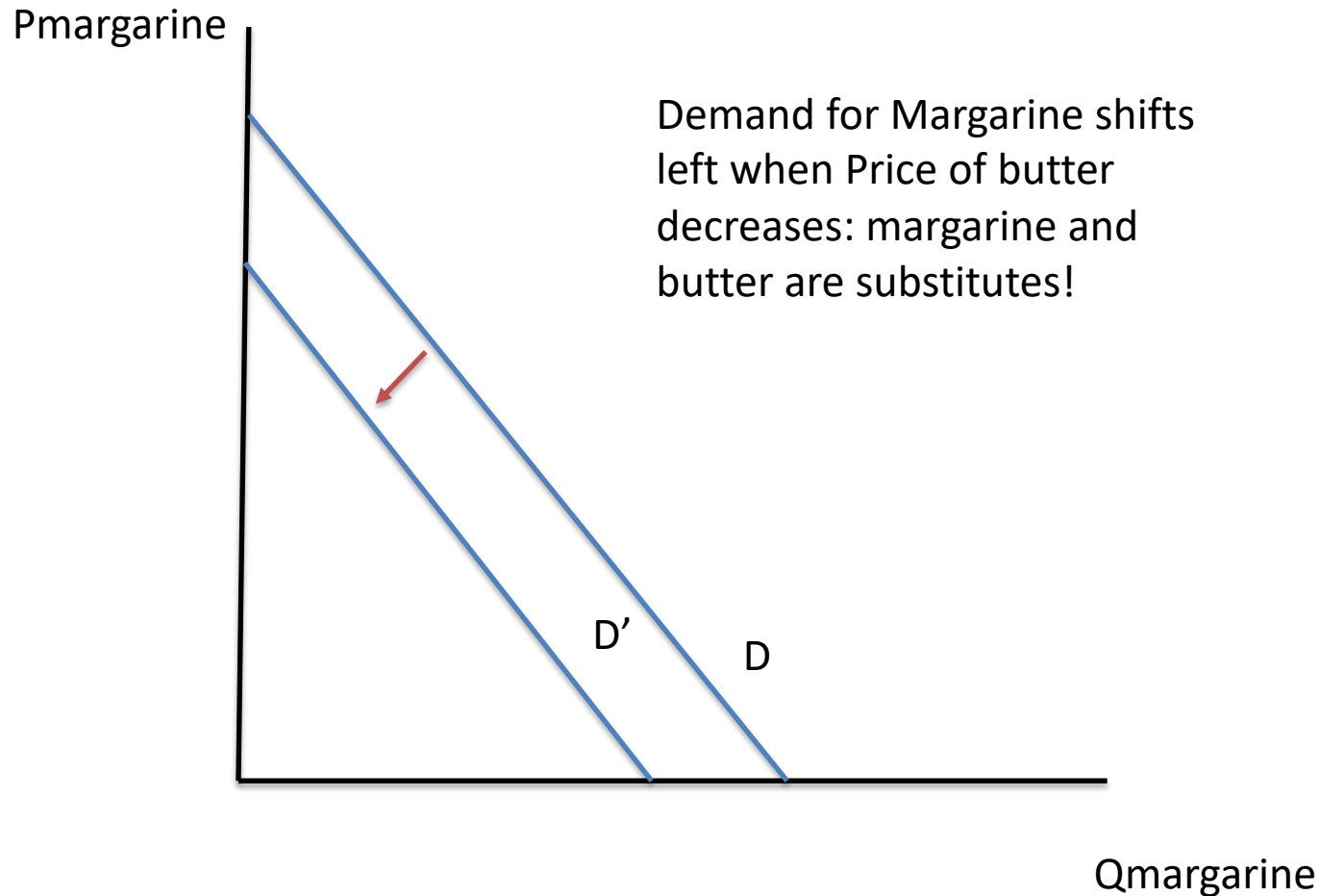
$$\begin{cases} \frac{y}{4x} = p_x \\ p_x x + y = 300 \end{cases} \Rightarrow x = \frac{60}{p_x}, y = 240$$

Price Changes and Shifts in Demand

- A change in the price of one good can shift the demand for a second good
- For example, a decrease in the price of butter (L_1 to L_3) causes a decrease in the quantity demanded of margarine without a change in the price of margarine
- Thus, in this example a decrease in the price of butter causes the demand for margarine to shift to the left



Price Changes and Shifts in Demand

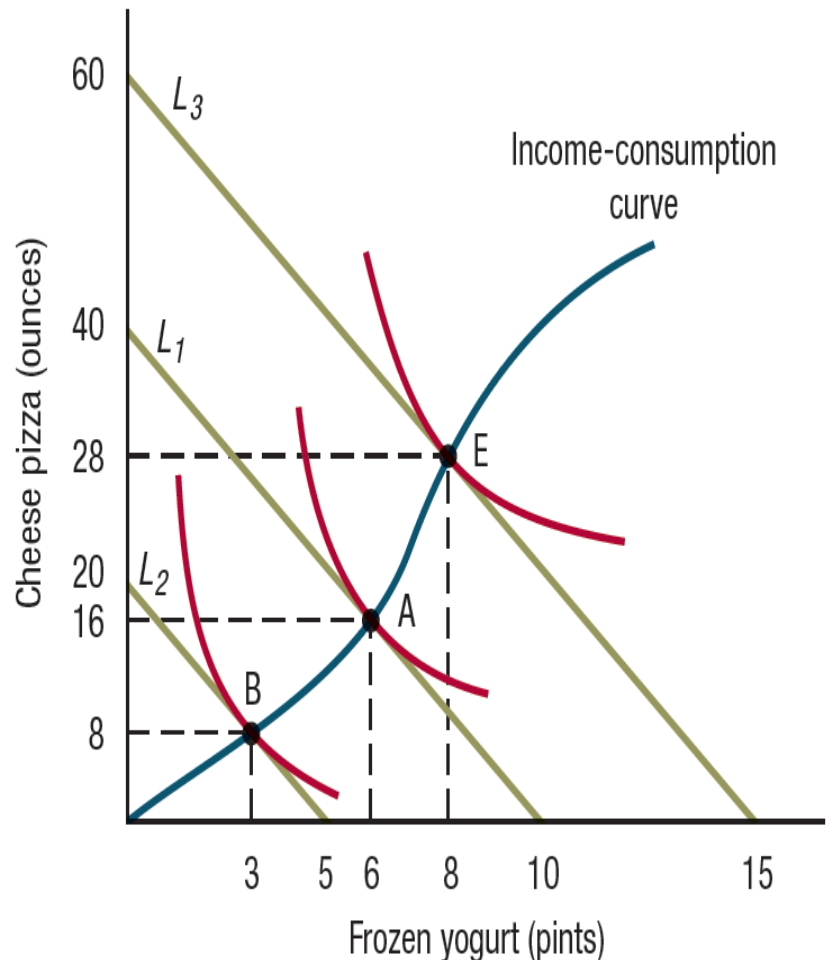


Price Changes and Shifts in Demand

- The effect of the change in the price of one good on the demand for a second good will depend on whether they are substitutes or complements (or neither)
- **Substitutes:** a decrease in P_1 leads to a leftward shift in the demand for P_2
- **Complements:** a decrease in P_1 leads to a rightward shift in the demand for P_2

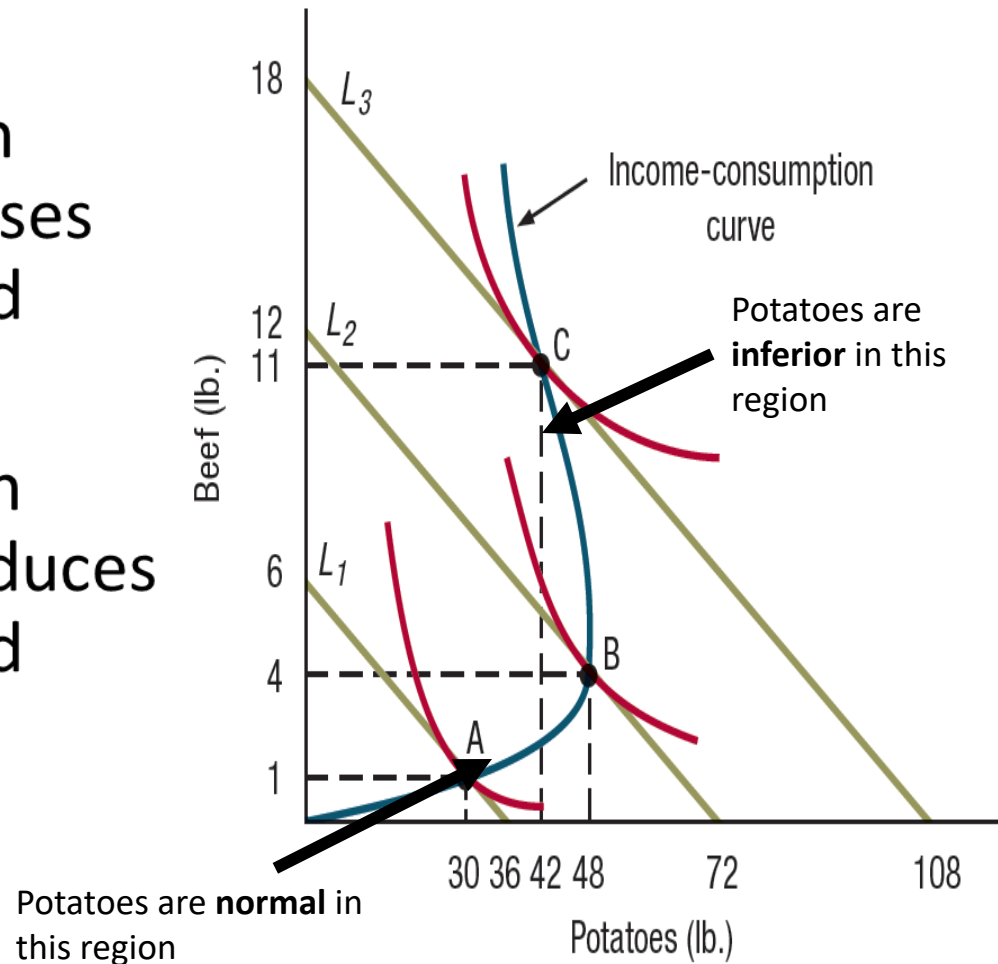
Income-consumption curve

- An **income effect** is the change in the consumption of a good that results from a change in income
- The **income-consumption curve** shows how the best affordable consumption bundle changes as income changes, holding everything else fixed



Normal vs Inferior goods

- If a good is **normal**, an increase in income raises the amount consumed ($E_M^d > 0$)
- If a good is **inferior**, an increase in income reduces the amount consumed ($E_M^d < 0$)



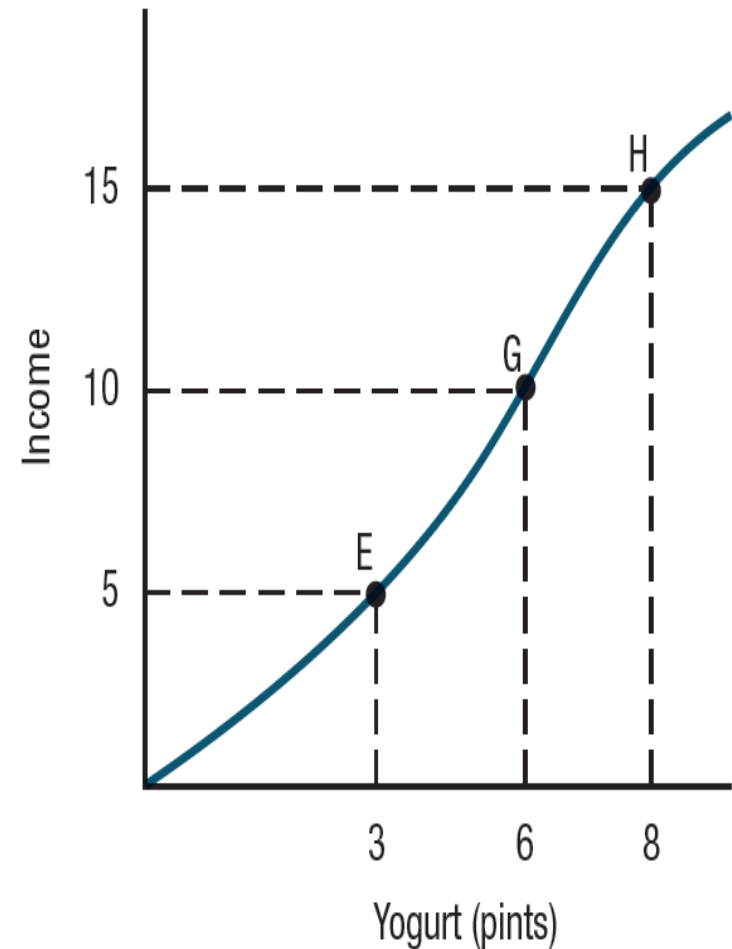
Engel Curve

- The **Engel curve** for a good describes the relationship between income and the amount consumed, holding everything else fixed
- Measure income on the vertical axis and amount consumed on the horizontal axis
- Engel curve slopes upward for a normal good and downward for an inferior one
- Derive Engel curve for X and Engel Curve for Y from income-consumption curve

Engel Curve

- In this example, yogurt is a normal good at all levels of income (upward sloping Engel curve)

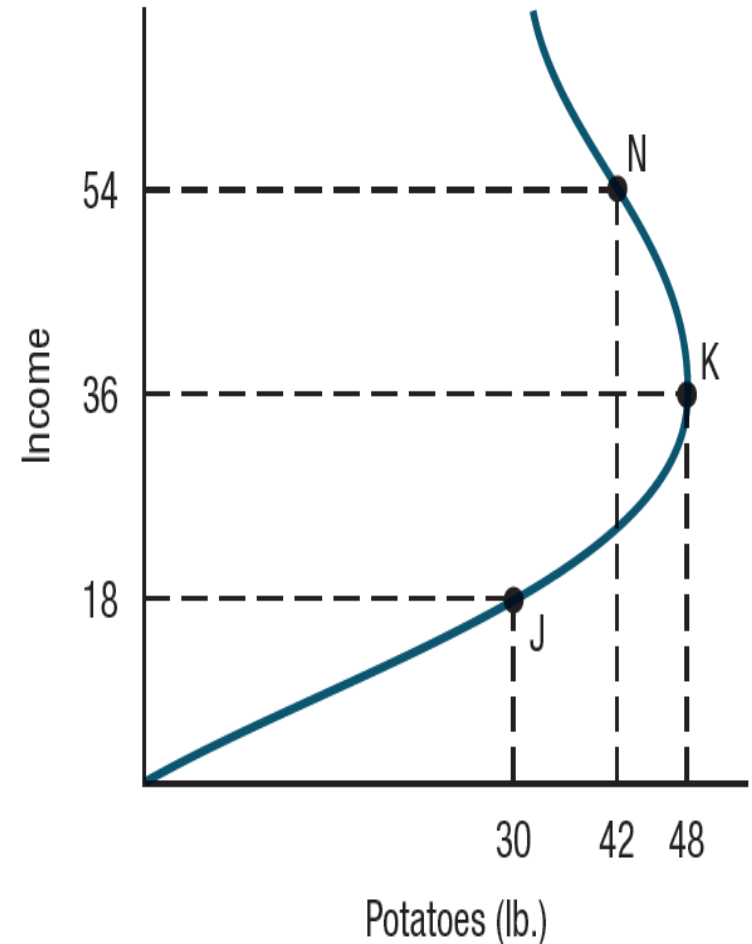
(a) Engel curve for yogurt



Engel Curve

- In this example, potatoes are normal when income is less than \$36 and inferior when income is greater than \$36

(b) Engel curve for potatoes



Back to our example

- Remember: we had $MRS_{xy} = (1/4)(y/x)$.
- Derive analytically the Engel curve for x and y as a function of M (still assuming that $p_y = 1$ € and $p_x = 0.4$).

$$\begin{cases} \frac{y}{4x} = 0.4 \\ 0.4x + y = M \end{cases} \Rightarrow x = \frac{M}{2}, y = 0.8M$$

Changes in Income and shifts of demand curve

- Demand curve shows relationship between price of a good and the amount purchased, holding everything else fixed, *including income*
- If income changes, the demand curve shifts
- If the good is normal
 - Income increase raises consumption at every price, so demand shifts to the right
 - Income decrease shifts demand to the left
- If the good is inferior, the effects are reversed

